Thesis Talk

Salome Bachmann

August 4, 2026

Overview of Topics

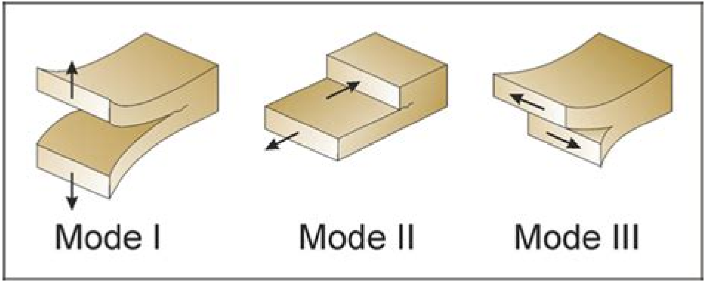
- Context
- Terminology
- Overview of Models
- Results of Cohesive Zone Model
- Disucussion
- Outlook

What am I doing and why is this relevant?

- Hang out in my office and have coffee (please come by)
- At the beginning, oversample and knit an entire top because the run time was too long
- Trying to make an accurate model for crack propagation in snow using SALVUS (FE)
- Cracks in snow on large scales can lead to avalanche release
- Average of 24 deaths by avalanche in Switzerland per year, multiple hundred non-fatal accidents (**LongtermStatistics**)
- Understanding relevant because avalanches endanger human life and infrastructure (**schweizerSnowAvalancheFormation2003**)
- **Ultimate Goal:** determining whether slab breakoff can be seen in DAS data and how this would look

Words I will propably say a lot explained I

- Slab: Part of the snowpack which breaks off and slides downhill as a cohesive unit (the avalanche itself, I'm looking at dry-snow slab avalanches) (**schweizerSnowAvalancheFormation2003**)
- Weak layer: Layer with different mechanical properties than the rest of the snow, cracks usually propagate in this layer because it's weaker than the rest of the snowpack (**schweizerSnowAvalancheFormation2003**)
- Mode I, Mode II and Mode III cracks: cracks opening as a response to stress in different directions:



Words I will propably say a lot explained II

Figure: Fracture modes according to **philippEffectsMechanicalLayering2013**

- Anticrack: weak layer collapses under loading, slab starts bending, collapse and cracks caused by bending propagate (**gaumeDynamicAnticrackPropagation2018**)
- Sub-Rayleigh: Cracks that propagate at speeds lower than the shear wave speed in snow (**trottetTransitionSubRayleighAnticrack2022**)
- Supershear: Cracks propagate faster than shear wave speed in snow (**trottetTransitionSubRayleighAnticrack2022**)

Overview of Models

- Model domain 400m wide, 3m high
- Absorbing boundary conditions 30m thick along top and both sides of domain
- No ABC at bottom because of theoretical snow interface
- 1.5m snow layer, 1.5m air layer
- Multiple source versions: single source, source line with sources firing with a time delay (moving source) and rupture front
- Sources buried in snow (like crack)
- For moving source: sub-rayleigh and supershear speeds
- 10Hz central frequency ricker source wavelet

My latest creation: the cohesive zone model (CZM)

- Model based on the cohesive zone model by **mahajanModelingInterfacialCrack2008**
- In CZM, fracturing is a gradual process, separation of the crack faces takes place over an extended crack tip, the cohesive zone (**mahajanModelingInterfacialCrack2008**)
- Resistance to opening by cohesive traction (**mahajanModelingInterfacialCrack2008**)
- In other words: there is a zone ahead of the crack tip, which already has some damage of the bonds
- Cracks represented as a fraction between different crack modes (Mode I-III)
- Caused by a a fault zone: starts at x=30m and ends at x=270m, 0.5m depth (middle of snow layer)
- Sources every 10cm firing with a time delay (dependend on supershear vs sub-rayleigh)

Supershear vs Sub-Rayleigh Space-Time Plots - Velocity

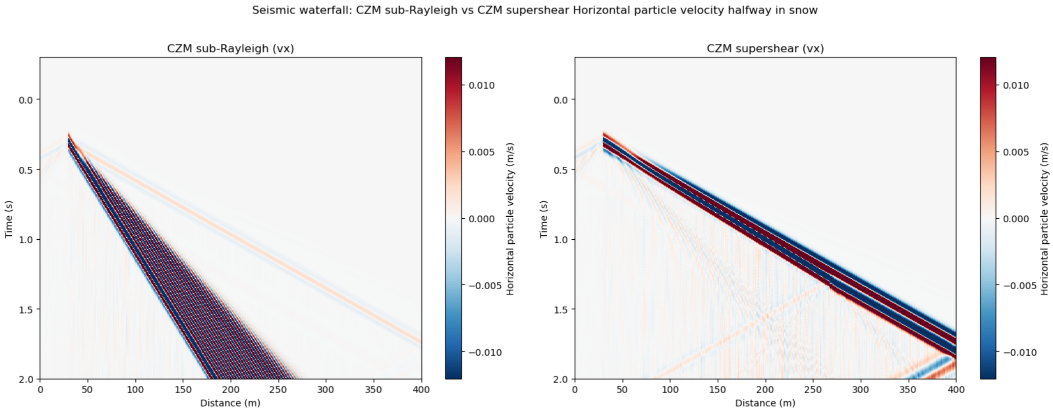


Figure: Horizontal Particle Velocity (vx) in sub-rayleigh (left) and supershear case (right)

Supershear vs Sub-Rayleigh Space-Time Plots - Strain

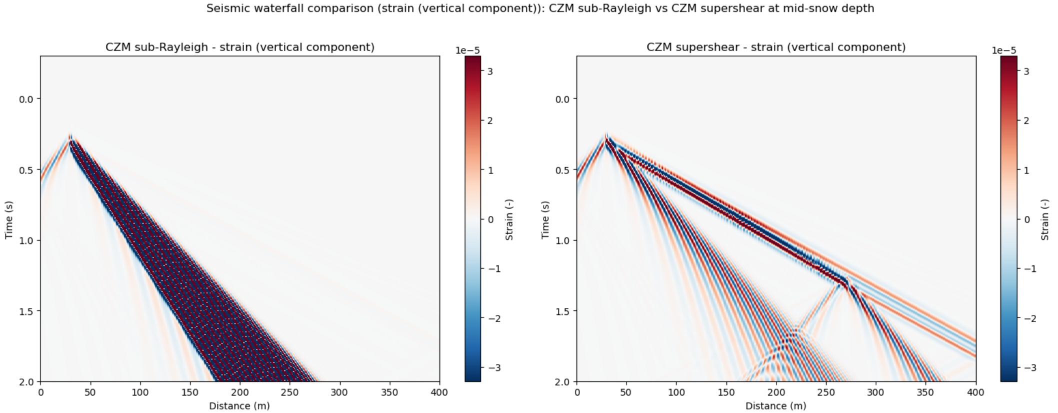


Figure: Strain in sub-rayleigh (left) and supershear case (right)

Possible interpretations of space-time plots I

- Sub-Rayleigh and Supershear case look very different (yay!)
- Blue-red-blue arrival before the main wavefront could be flexural wave (bending of the slab near the source)
- Blue first: negative velocity –> source in negative y (slab is pushed down by the source)
- Red next: because of elastic rebound of slab
- Blue again: oscillation in source direction after rebound
- This can be seen in vx for sub-rayleigh and strain for supershear
- In sub-rayleigh, waves travel slower than the flexural waves, which is very noticeable when looking at vx
- In supershear vx, rupture arrives with flexural waves
- Flexural waves can be seen in strain supershear plot: because strain is spatial derivative, which acts as a high-pass and amplifies flexural waves

Possible interpretations of space-time plots II

- This is because the derivative acts as a multiplication with $j\omega$, which is dependent on frequency: the higher the frequency, the greater $|j\omega|$, which means there is amplification of high frequencies (**oppenheimSignalsSystems1997**)
- Supershear case: Strong first arrivals, weak arrivals at later times
- "Reflection" seen in bottom rigth corner is rupture arrest front (energy could increase because rupture stops at a point)
- This is only visible in supershear case because there is constructive interference between the rupture front and the wavefronts because they move at similar speeds in supershear

Flexural wave -> early warning system?

- Because flexural wave arrives before main wavefront of cracks: could be used as early warning system
- Verify by picking a reciever and picking arrival between flexural and main wavefront

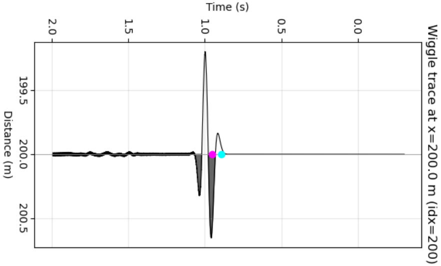


Figure: Arrivals for trace at x=200m, teal is first arrival (precursor), pink is main arrival

- Difference in arrivals 0.058s in supershear case: not very early warning, but still a difference

Much needed sanity checks I

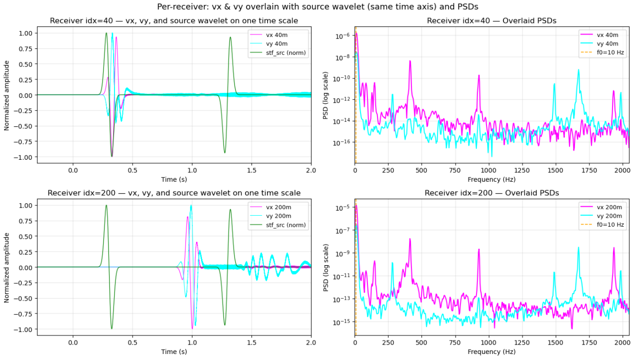
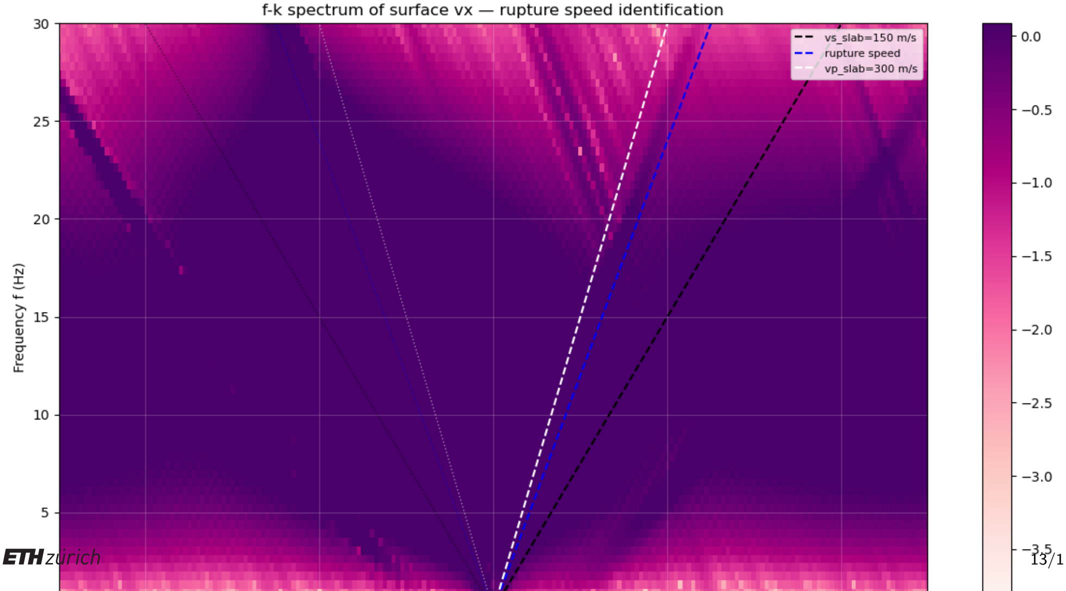


Figure: Traces and power spectral density (PSD) at recievers at 40m and 200m. Vertical velocity in blue, horizontal velocity in pink, source trace in green

Much needed sanity checks II



Was it all just a dream?

- Source wavelet pulse (green) matches traces at both reciever locations
- Far trace (200m) has stronger coda noise in vy: could be rupture arrest?
- All PSD have a clear peak at the source frequency (10Hz)
- F-k plot shows high energies at same slope of rupture speed: model takes the correct speed
- Trailing (s) wave can also be seen (line with similar slope to black line)
- Waves propagate in both directions (in rupture direction as well as against it)
- Probably not a dream, my model might be correct!

Next Steps

- Add weak layer to model
- Add topography to model
- Make an anticrack model using the Material Point Method (MPM)
- ???

Thank you for your attention! Any Questions or Suggestions?

Bibliography I

Salome Bachmann
Master Student
sbachmann@student.ethz.ch

ETH Zurich
D-EAPS